

Technical Manual

Metal Spinning CNC Controller

Documentation

Technical Manual - Metal Spinning CNC Controller

For System Owner / Technical Lead

1. System Architecture

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The system follows a hierarchical structure:

- * OB1 calls FB_Process every scan cycle
- * FB_Process orchestrates all sub-systems
- * Supporting FBs handle specific functions (inputs, safety, limits, alarms)
- * Operation FBs execute motion (G-code, tool change, spindle)
- * Motion FBs wrap Siemens MC_ instructions

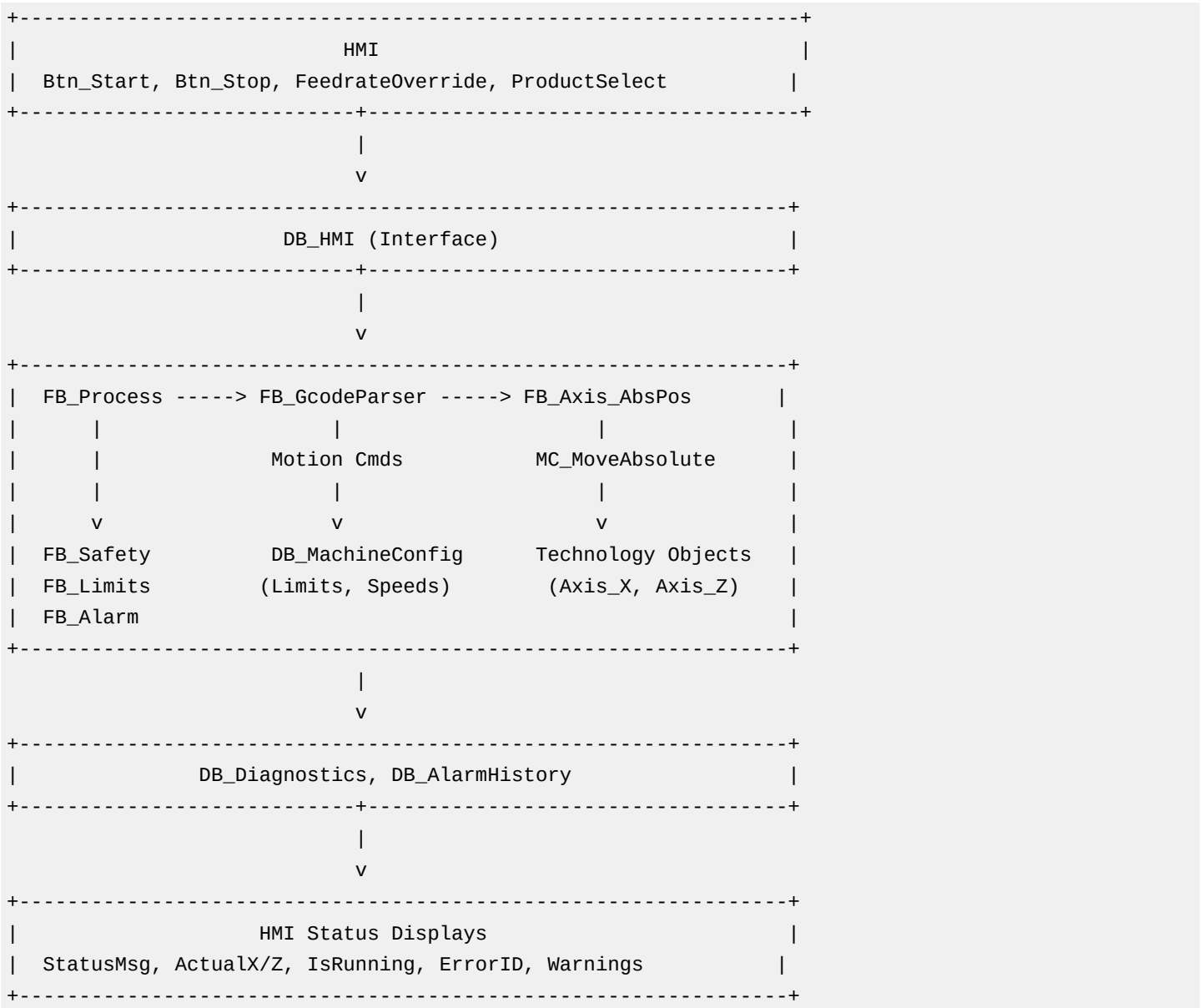
2. Main State Machine

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State	ID	Description
STOPPED	0	Idle, waiting for Start
STARTING	10	Checking if homing needed
HOMING	15	Referencing axes
RUNNING	20	Executing G-code
PAUSED	25	Halted, waiting for Resum
TOOL_CHANGE	30	Initiating tool change
TOOL_WAIT	35	Waiting for tool changer
ERROR	999	Error active, needs Reset

3. File Structure

File	Purpose	Size
`01_DataTypes.scl`	UDT_RecipeData	3 KB



5. Error Code Reference

Code	Category	Message
0x0001	Motion	X Axis Move Failed
0x0002	Motion	Z Axis Move Failed
0x0003	Motion	X Axis Homing Failed
0x0004	Motion	Z Axis Homing Failed
0x0008	Motion	Motion Timeout
0x0101	Limits	X Below Soft Limit
0x0102	Limits	X Above Soft Limit
0x0103	Limits	Z Below Soft Limit
0x0104	Limits	Z Above Soft Limit
0x0111	Limits	X Hit Min Limit Switch
0x0112	Limits	X Hit Max Limit Switch
0x0113	Limits	Z Hit Min Limit Switch
0x0114	Limits	Z Hit Max Limit Switch
0x0203	Tool	Tool Change Failed
0x0305	G-code	Target Outside Limits
0x0401	Safety	EMERGENCY STOP
0x0402	Safety	Safety Door Open
0x0404	Safety	Air Pressure Low
0x0501	Spindle	Spindle Drive Fault
0x0502	Spindle	Spindle Velocity Error

6. Linear Interpolation

Both axes finish simultaneously using velocity scaling:

```
Distance:  distTotal = sqrt(deltaX2 + deltaZ2)
Time:      moveTime = distTotal / feedrate
Velocities: velX = deltaX / moveTime
           velZ = deltaZ / moveTime
```

7. Technology Objects

Name	Type	Purpose
Axis_X	TO_PositioningAxis	Radial motion

Axis_Z	TO_PositioningAxis	Axial motion
Axis_Tool	TO_PositioningAxis	Tool turret
Axis_Spindle	TO_PositioningAxis	VFD via PTO

8. Key Design Decisions

Decision	Rationale
Separate text/recipe hand	Different input sources,
Modal feedrate	Match standard G-code beh
Safety checked every scan	Never skip, even during e
Timeouts on all waits	Prevent indefinite hangs
Soft limits before move	Reject invalid targets ea